

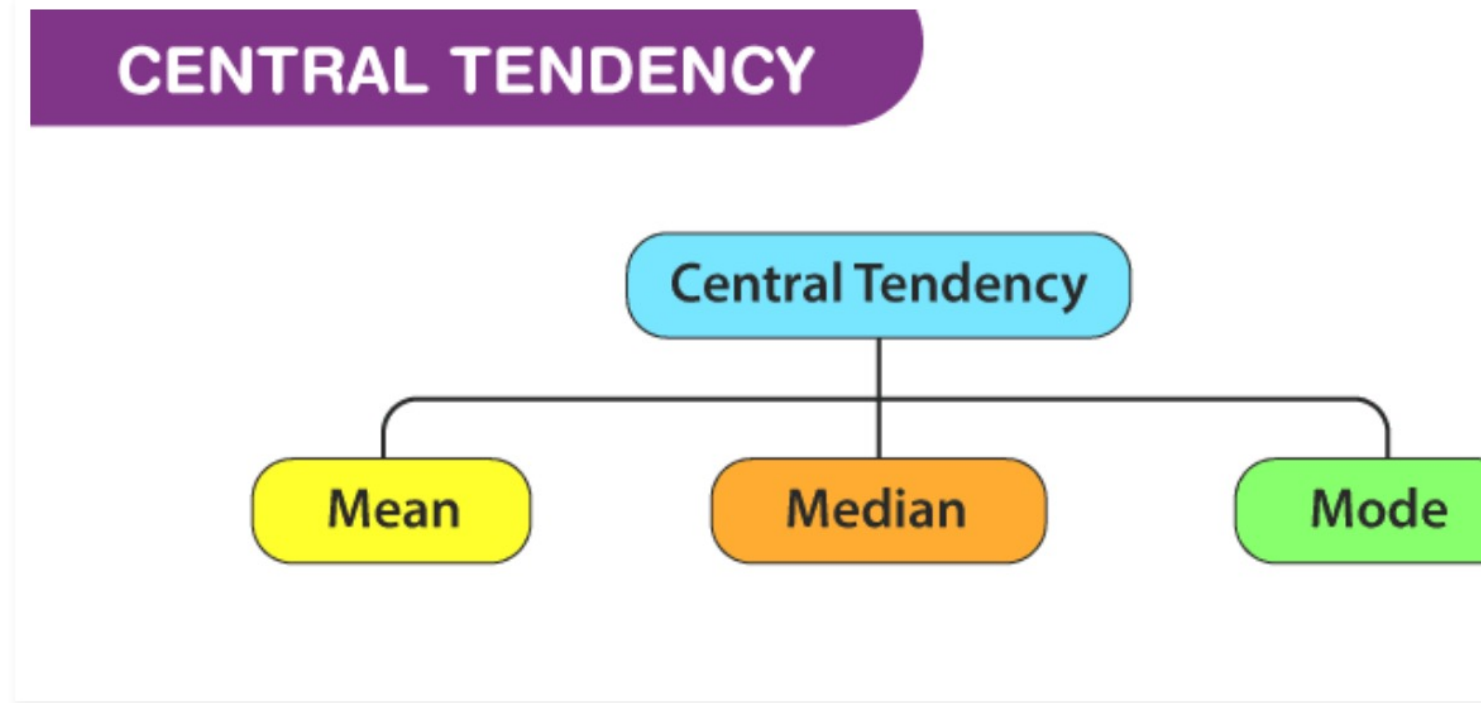
Central Tendency

In statistics, the central tendency is the descriptive summary of a data set. Through the single value from the dataset, it reflects the centre of the data distribution.

Definition:

The central tendency is stated as the statistical measure that represents the single value of the entire distribution or a dataset. It aims to provide an accurate description of the entire data in the distribution.

Measures of Central Tendency



What is the MEAN?
How do we find it?



The mean is the numerical average of the data set.

The mean is found by adding all the values in the set, then dividing the sum by the number of values.

Mean = (Sum of all the observations/Total number of observations)

Example:

What is the mean of 2, 4, 6, 8 and 10?

Solution:

First, add all the numbers.

$$2 + 4 + 6 + 8 + 10 = 30$$

Now divide by 5 (total number of observations).

$$\text{Mean} = 30/5 = 6$$

In the case of a discrete probability distribution of a random variable X , the mean is equal to the sum over every possible value weighted by the probability of that value; that is, it is computed by taking the product of each possible value x of X and its probability $P(x)$ and then adding all these products together.

The symbol of mean is usually given by the symbol 'x̄'. The bar above the letter x, represents the mean of x number of values.

$$\begin{aligned}\bar{X} &= (\text{Sum of values} \div \text{Number of values}) \\ &= (x_1 + x_2 + x_3 + \dots + x_n) / n\end{aligned}$$

Ungrouped Data(individual series)

In a class there are 20 students
and they have secured a percentage of

88

82

88

85

84

80

81

82

83

85

84

74

75

76

89

90

89

80

82

83

Find the mean percentage obtained by the class.

Solution:

Mean = Total of percentage obtained by 20 students in
class/Total number of students

$$= [88 + 82 + 88 + 85 + 84 + 80 + 81 + 82 + 83 + 85 + 84 + 74 + 75 + 76 + 89 + 90 + 89 + 80 + 82 + 83]/20$$
$$= 1660/20$$

$$= 83$$

Hence, the mean percentage of each student in the class
is 83%.

There are 3 methods of calculating arithmetic mean in the case of individual series , discrete series:

✓ **1. Direct Method**

2. Assumed Mean Method

3. Step-deviation method

Students of class Vth got the following marks in their statistics paper. Calculate arithmetic mean .

X(MARKS)

- 20
- 44
- 65
- 28
- 45
- 67
- 30
- 50
- 68
- 39
- 53
- 70
- 40
- 60
- 75

Test yourself

Discreet series

Example 1. Calculate A.M. from the following data

Marks obtained :	4	8	12	16	20
No. of students :	6	12	18	15	9

Solution:

Marks X	No. of students f	fX
4	6	24
8	12	96
12	18	216
16	15	240
20	9	180
	$N = 60$	$\Sigma fX = 756$

$$\begin{aligned}\text{As } \bar{X} &= \frac{\Sigma fX}{N} \\ \therefore \bar{X} &= \frac{756}{60} = 12.6\end{aligned}$$

Calculate arithmetic mean of following data .

Size	30	29	28	27	26	25	24	23	22	21
Frequency (f)	2	4	5	3	2	7	1	4	5	7

Test yourself

Continuous series

Direct Method

Example 1. Use direct method to find $\bar{X}$.

Income :	10-20	20-30	30-40	40-50	50-60	60-70
No. of Persons :	4	7	16	20	15	8

Solution

Income	Mid value m	No. of Persons f	fm
10-20	15	4	60
20-30	25	7	175
30-40	35	16	560
40-50	45	20	900
50-60	55	15	825
60-70	65	8	520
		$N = 70$	$\Sigma fm = 3040$

$$\text{As } \bar{X} = \frac{\Sigma fm}{N}$$

$$\therefore \bar{X} = \frac{3040}{70} = 43.43.$$

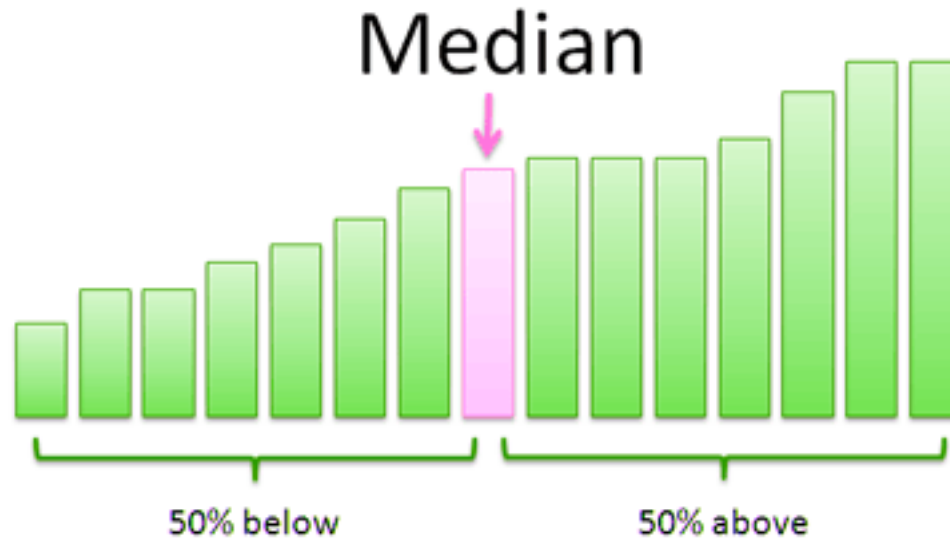
Calculate arithmetic mean of following data .

C.I. :	10–20	20–30	30–40	40–50	50–60
f :	3	7	11	9	6

Test yourself ???

Median

Median, in statistics, is the middle value of the given list of data, when arranged in an order. The arrangement of data or observations can be done either in ascending order or descending order.



Median when count of number is odd

Example: 3, 13, 7, 5, 21, 23, 39, 23, 40, 23, 14, 12, 56, 23, 29

When we put those numbers in order we have:

3, 5, 7, 12, 13, 14, 21, 23, 23, 23, 23, 29, 39, 40, 56

There are **fifteen** numbers. Our middle is the **eighth** number:

3, 5, 7, 12, 13, 14, 21, **23**, 23, 23, 23, 29, 39, 40, 56

The median value of this set of numbers is **23**.

$$M = n + 1/2$$

Total number of observations are

odd i.e; 15

$15 + 1/2 = 8^{\text{th}}$ element of sorted list is median

$$M = 23$$

Median when count of number is even

Example: 3, 13, 7, 5, 21, 23, 23, 40, 23, 14, 12, 56, 23, 29

When we put those numbers in order we have:

3, 5, 7, 12, 13, 14, 21, 23, 23, 23, 23, 29, 40, 56

There are now **fourteen** numbers and so we don't have just one middle number, we have a **pair of middle numbers**:

3, 5, 7, 12, 13, 14, **21, 23**, 23, 23, 23, 29, 40, 56

In this example the middle numbers are **21 and 23**.

To find the value halfway between them, add them together and divide by 2:

$$\begin{aligned} 21 + 23 &= 44 \\ \text{then } 44 \div 2 &= 22 \end{aligned}$$

So the **Median** in this example is **22**.

$$M = n + 1/2$$

Total number of observations are **even** i.e; 14

$14 + 1/2 = 7.5^{\text{th}}$ element of sorted list is median

$$\text{Median} = 7^{\text{th}} \text{ term} + 8^{\text{th}} \text{ term} / 2$$

$$21 + 23 / 2 = 22$$

$$M = 22$$

Find Median ?

102, 56, 34, 99, 89, 101, 10, 77

Test yourself ???

Median in Discreet series

Example 1. Compute Median for following data

X :	10	20	30	40	50	60	70
f :	4	7	21	34	25	12	3

Solution

X	f	C _f	Terms
10	4	= 4	(1-4)
20	7	(4 + 7) = 11	(5-11)
30	21	(11 + 21) = 32	(12-32)
40	34	(32 + 34) = 66	(33-66)
50	25	(66 + 25) = 91	(67-91)
60	12	(91 + 12) = 103	(92-103)
70	3	(3 + 103) = 106	(104-106)
	N = 106		N = 106

Now M = Size of $\left(\frac{N+1}{2}\right)$ th term = Size of $\left(\frac{106+1}{2}\right)$ th term = Size of 53.5th term = 40

Median = 40

SPECIAL CASE

Example 2. Find Median from the following data.

X :	4	8	12	16	20	24	28
f :	7	18	25	18	15	10	7

Solution

X	f	C _f	Terms
4	7	7	(1-7)
8	18	25	(8-25)
12	25	50	(26-50)
16	18	68	(51-68)
20	15	83	(69-83)
24	10	93	(84-93)
28	7	100	(94-100)
	N = 100		

$$\begin{aligned}
 \text{Median} &= \text{Size of } \left(\frac{n+1}{2}\right) \text{th term} \\
 &= \frac{100+1}{2} \text{th term} = 50.5 \text{th term} \\
 &= \frac{50\text{th} + 51\text{st}}{2} \text{ term} = \frac{12 + 16}{2} = 14.
 \end{aligned}$$

Median = 14

Find Median ?

X :	8	12	20	25	30	40
f :	9	16	28	46	20	10

Test yourself ???

Median in Continuous Series:

Class Interval :	0-10	10-20	20-30	30-40	40-50	50-60	60-70
Frequency :	7	18	34	50	35	20	6

Solution

Class Interval	f	Cf
0-10	7	7
10-20	18	25
Cf 20-30	34	59
f 30-40	50	109
40-50	35	144
50-60	20	164
60-70	6	170
	N = 170	

$$N_1 = \frac{N}{2} = \frac{170}{2} = 85, ; C_f = 59 ; f = 50; L = 30$$

$$\text{As } M = L + \frac{N_1 - C.f.}{f} \times i$$

$$\therefore M = 30 + \frac{85 - 59}{50} \times 10$$

$$= 30 + \frac{26}{5} = 30 + 5.2 = 35.2$$

2nd Formula (L as upper limit) can also be used. But we often use first Formula.

$$M = L - \frac{C_f - N_1}{f} \times i$$

Here $C_f = 109$ should be taken ; And L is upper limit.

$$\therefore M = 40 - \frac{109 - 85}{50} \times 10 = 40 - \frac{24}{5} = 40 - 4.8 = 35.2.$$

Calculate the median for an exclusive (continuous) distribution given below.

Class Interval	Frequency
10 – 20	11
20 – 30	13
30 – 40	13
40 – 50	9
50 – 60	4

Test yourself ???

Solution

Class Interval	Frequency (f)	Cumulative frequency (f_c)
10 – 20	11	11
20 – 30	13	24
30 – 40	13	37
40 – 50	9	46
50 – 60	4	50

Here we see that the total number of observation is $N = 50$. Therefore (30 – 40) is the median class. We also observe that $LRL = 30$, $f_c = 24$, $f_m = 13$ and $i = 20 - 10 = 10$. Now we are in a position to use the formula for median:

$$\text{median} = LRL = \frac{\left(\frac{N}{2}\right) - f_x}{f_m} \times i = 30 + \frac{(25 - 24)}{13} \times 10 = 30 + \frac{10}{13} = 30.77$$

approximately.

Less than case !

Marks	Number of Students
Less than 10	12
Less than 20	26
Less than 30	40
Less than 40	58
Less than 50	80
Less than 60	110
Less than 70	138
Less than 80	150

Marks	CF	F Step:2
0 – 10	12	12
10 – 20	26	26-12 = 14
20 – 30	40	40-26 = 14
30 – 40	58	58-40 = 18
40 – 50	80	80-58 = 22
50 – 60	110	110-80 = 30
60 – 70	138	138-110 = 28
70 – 80	150	150-138 = 12

Marks	Number of Students
0 – 10	12
10 – 20	26
20 – 30	40
30 – 40	58
40 – 50	80
50 – 60	110
60 – 70	138
70 – 80	150

Step 1:

Step 3: Find the **median class** using $N/2$

Where N is the sum of frequency. Median Class = $150/2 = 75$.

Locate 75 ($N/2$) in cumulative frequency and pick the next number available in cf after 75

$$\text{Median} = 40 + \{(75 - 58) * 10\} / 22$$

$$\text{Median} = 40 + 170/22$$

$$\text{Median} = 40 + 7.727$$

$$\text{Median} = 47.727 \text{ or } 47.73 \text{ (Approx)}$$

$$M = \text{lower limit} + \frac{N/2 - cf^p}{f^e} \times i$$

Calculate the median from the following data:

Value	Frequency
Less than 10	4
Less than 20	16
Less than 30	40
Less than 40	76
Less than 50	96
Less than 60	112
Less than 70	120
Less than 80	125

Test yourself ???

Solution

Forming the Cumulative frequency table we have.

Value	Frequency	C.f
0-10	4	4
10-20	12	16
20-30	24	40
30-40	36	76
40-50	20	96
50-60	16	112
60-70	8	120
70-80	5	125

Here middle term is $\left(\frac{125}{2}\right)^{\text{th}}$ or 62.5th observation, which lies in class 30-40.

Thus,, we have

Median class = 30 - 40

$$l = 30, \quad \frac{n}{2} = 62.5$$

c.f. = 40, $f = 36$ and $h = 10$

Putting these values in the formula of median

$$\begin{aligned}\text{Median} &= l + \left[\frac{\frac{n}{2} - \text{c.f.}}{f} \right] \times h \\ &= 30 + \left[\frac{62.5 - 40}{36} \right] \times 10 \\ &= 30 + \frac{22.5}{36} \times 10\end{aligned}$$

$$= 30 + 6.25 = 36.25$$

Hence, median = 36.25

More than case !

Age greater than (Years)	Number of Persons
More than 0	230
More than 10	218
More than 20	200
More than 30	165
More than 40	123
More than 50	73
More than 60	28
More than 70	8

Class Interval	Cumulative Frequency (greater than)	Frequency	Cumulative Frequency (less than)
0-10	230	230 – 218 = 12	12
10-20	218	218 – 200 = 18	30
20-30	200	200 – 165 = 35	65
30-40	165	165 – 123 = 42	107
40-50	123	123 – 73 = 50	157
50-60	73	73 – 28 = 45	202
60-70	28	28 – 8 = 20	222
70+	8	8	230

$$\Rightarrow \text{Median} = 40 + \left(\frac{115 - 107}{50} \right) \times 10$$

$$\Rightarrow \text{Median} = 40 + \left(\frac{8}{50} \right) \times 10 \quad \text{Median}=41.6$$

$$\text{Median} = l + \frac{\frac{N}{2} - C}{f} \times h$$

Marks (more than or equal to)	Number of students
More than 80	150
More than 90	141
More than 100	124
More than 110	105
More than 120	60
More than 130	27
More than 140	12
More than 150	0

Test yourself ???

Solution:

Marks	No. of students	Cumulative frequency
80 - 90	$150 - 141 = 9$	9
90 - 100	$141 - 124 = 17$	$9 + 17 = 26$
100 - 110	$124 - 105 = 19$	$26 + 19 = 45$
110 - 120	$105 - 60 = 45$	$45 + 45 = 90$
120 - 130	$60 - 27 = 33$	$90 + 33 = 123$
130 - 140	$27 - 12 = 15$	$123 + 15 = 138$
140 - 150	$12 - 0 = 12$	$138 + 12 = 150$

$$\text{Here, } n = 150 \Rightarrow \frac{N}{2} = \frac{150}{2} = 75$$

The c.f. just greater than 75 is 90 and the corresponding class is 110 - 120.

Hence, 110 - 120 is the median class.

$$\therefore \ell = 110, f = 45, cf = 45, h = 10$$

$$\therefore \text{Median} = \ell + \left(\frac{\frac{N}{2} - cf}{f} \right) \times h$$

$$= 110 + \left(\frac{75 - 45}{45} \right) \times 10$$

$$= 110 + \frac{300}{45} = 110 + 6.67 = 116.67$$